Quant Researcher Interview Playbook

Time Series, Signals & Statistical Research

A Desk2Quant Professional Guide for Quant Research Interviews, Signal Design, Validation, and Portfolio Construction

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What this playbook is

This is not a catalogue of statistical definitions. It is a research operating manual: how to transform noisy market data into falsifiable hypotheses, estimate signals without leakage, measure decay and breadth, survive multiple testing, validate through time, construct portfolios under costs and constraints, and defend the entire chain in a quant researcher interview.

How to Use This Book

Who This Is For

1. Quant Research candidates: Use the book as an interview map. You should be able to move from data-generating process → hypothesis → estimator → validation → portfolio → post-trade diagnosis without changing vocabulary mid-answer.
2. Quant traders moving toward research: Prioritize signal decay, cross-sectional ranking, capacity, transaction costs, and the distinction between statistically significant and economically tradable effects.
3. Quant developers / research engineers: Prioritize point-in-time data, reproducibility, walk-forward infrastructure, feature availability, and the implementation checklists.

The Five Ways Research Interviews Go Wrong

1. You quote a model but never state the data-generating assumptions.
2. You report a t-statistic but ignore serial/cross-sectional dependence.
3. You show an in-sample Sharpe but cannot explain the search process that produced it.
4. You cross-validate randomly even though the labels overlap in time.
5. You optimize a portfolio before proving the signal survives costs, decay, and regime shifts.

The Research Chain

A strong answer follows one chain: **Question** → **Data** → **Timing** → **Hypothesis** → **Estimation** → **Falsification** → **OOS Validation** → **Costs** → **Portfolio** → **Monitoring**. If one link is vague, the research is not production-ready.

The Learning Protocol

Each module presents core ideas through six layers so that readers can build from first principles to interview-level fluency.

1. **Plain-English intuition:** what problem the concept solves and what mental picture to keep.
2. **Equation:** the smallest mathematical representation that captures the idea.
3. **Symbol-by-symbol meaning:** what every variable means before manipulating the formula.
4. **Direct proportional relationships:** what increases, what decreases, and what scales with time, sample size, volatility, turnover, or breadth.
5. **Tiny numerical example:** a calculation small enough to do mentally in an interview.
6. **Mnemonic:** a compressed memory device for recall under pressure.

The goal is not formula memorization. The goal is to be able to say: *"I know what this measures, why it exists, how it scales, when it breaks, and what I would do next."*

Contents

Module 1: The Quant Research Operating System

From idea generation to falsification

Module objective. Build a disciplined research process that separates a clever story from reproducible evidence.

What You Must Be Able to Do

By the end of this module, you should be able to define the object mathematically, explain the research intuition, identify at least two failure modes, describe a time-respecting implementation, and answer the associated interview questions without hiding behind software output.

Module map:

- 1.1 Research Is a Sequence of Conditional Claims
- 1.2 The Research Hypothesis Ledger
- 1.3 Reproducibility as a Research Skill
- 1.4 Interview Drill: Defend a Research Project

Desk2Quant Standard

For every concept, connect five layers: **equation** → **assumption** → **data timing** → **failure mode** → **portfolio consequence**. This is the difference between knowing statistics and doing quant research.

1.0 Beginner First-Principles Primer

Quant research is disciplined uncertainty reduction. Before learning ARIMA, GARCH, PCA, or portfolio optimization, you need a mental operating system for turning an idea into a falsifiable experiment.

How to study this module if you are new

Read each concept in this order: **intuition first, equation second, proportional relationships third, numerical example fourth, mnemonic last**. Do not memorize the equation until you can explain what would happen if each important input doubled, halved, or approached a boundary.

Concept 1: Hypothesis, Null, and Alternative

Plain-English intuition. A research idea becomes testable only when you specify what would count as failure. The null hypothesis is the boring world in which your feature adds no predictive information; the alternative is the world in which it does. This prevents you from changing the story after seeing the result.

Core equation.

$$H_0: \ \mathbb{E}[r_{t+h} \mid x_t] = \mathbb{E}[r_{t+h}], \qquad H_1: \ \mathbb{E}[r_{t+h} \mid x_t] \neq \mathbb{E}[r_{t+h}].$$

What the symbols mean. Here x_t is the information available at decision time t, r_{t+h} is the future return over horizon h, and $\mathbb{E}[\cdot]$ denotes an expectation. The conditioning bar "|" means "given what I know now."

Direct Proportional Relationships

- If the true effect size grows, statistical power generally increases: Power $\uparrow$ as $|\beta| \uparrow$.
- For IID noise, the standard error decreases approximately as $1/\sqrt{N}$. Equivalently, inverse-standard-error precision and signal-to-noise evidence scale approximately as $\sqrt{N}$.
- The more researcher degrees of freedom you introduce, the greater the probability of finding a false winner.

Tiny worked example. Suppose the unconditional next-month return is 0.5%, but stocks in the top signal decile average 0.8%. The research question is whether the extra 0.3% is stable and distinguishable from sampling noise, not whether one backtest line slopes upward.

Memory Anchor

Mnemonic: HNA = Hypothesis, Null, Alternative. Never start with a model; start with the claim you are willing to kill.

Concept 2: Decision Timestamp and Information Set

Plain-English intuition. Every signal lives on a clock. A feature is valid only if it was knowable before the trade decision. Quant research therefore begins by defining the information set, not by choosing the regression.

Core equation.

$$\mathcal{I}_t = \{\text{all information actually available by time } t\}, \qquad x_t \in \mathcal{I}_t.$$

What the symbols mean. $\mathcal{I}_t$ is the information set at time t. A valid feature x_t must be measurable with respect to that set. A revised accounting value published next month is not in $\mathcal{I}_t$ today.

Direct Proportional Relationships

- Leakage risk increases with publication/revision latency.
- As execution delay increases, usable alpha usually decreases when signals decay quickly.
- Shorter horizons make timestamp precision more important because a few minutes can represent a large fraction of the holding period.

Tiny worked example. If earnings are released at 16:05, a strategy trading at the 16:00 close cannot use that earnings surprise. The same value becomes legal for the next session.

Memory Anchor

Mnemonic: KAT = Knowable At Time. If you cannot prove KAT, do not backtest it.

Module 8: Hypothesis Testing Under Dependence

t-stats that mean what you think they mean

Module objective. Use inference that respects serial correlation, heteroskedasticity, cross-sectional dependence, and overlapping labels.

What You Must Be Able to Do

By the end of this module, you should be able to define the object mathematically, explain the research intuition, identify at least two failure modes, describe a time-respecting implementation, and answer the associated interview questions without hiding behind software output.

Module map:

- 8.1 The t-Statistic Is a Model
- 8.2 Newey–West / HAC
- 8.3 Bootstrap and Permutation Tests
- 8.4 Interview Drill: Significant but Unconvincing

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8.0 Beginner First-Principles Primer

Inference asks whether an observed effect is larger than what noise could plausibly produce. In finance, dependence and heavy tails make textbook IID standard errors dangerous.

How to study this module if you are new

Read each concept in this order: **intuition first, equation second, proportional relationships third, numerical example fourth, mnemonic last**. Do not memorize the equation until you can explain what would happen if each important input doubled, halved, or approached a boundary.

Concept 1: Estimate, Standard Error, and t-Statistic

Plain-English intuition. A coefficient alone says nothing about precision. The standard error measures sampling uncertainty; the t-statistic measures estimate size in standard-error units.

Core equation.

$$t = \frac{\hat{\beta} - \beta_0}{\mathrm{SE}(\hat{\beta})}.$$

Module 14: Walk-Forward Validation, Purging & Embargo

Time-respecting model selection

Module objective. Validate models the way they would have been selected through history, especially when labels overlap and hyperparameters evolve.

What You Must Be Able to Do

By the end of this module, you should be able to define the object mathematically, explain the research intuition, identify at least two failure modes, describe a time-respecting implementation, and answer the associated interview questions without hiding behind software output.

Module map:

- 14.1 Why Random Cross-Validation Fails
- 14.2 Purging and Embargo for Overlapping Labels
- 14.3 Nested Validation and Hyperparameters
- 14.4 Model Decay and Monitoring

Desk2Quant Standard

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14.0 Beginner First-Principles Primer

Validation must respect time. The question is not "can the model fit old data?" but "would this exact research process have survived repeated future deployments?"

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Concept 1: Chronological Train/Validation/Test Split

Plain-English intuition. Time-ordered data should remain time-ordered. Training learns parameters, validation chooses models, and test is a final untouched estimate of future-like performance.

Core equation.

$$\underbrace{[1,\ldots,T_{train}]}_{\text{fit}} \rightarrow \underbrace{[T_{train}+1,\ldots,T_{val}]}_{\text{select}} \rightarrow \underbrace{[T_{val}+1,\ldots,T]}_{\text{final test}}.$$

17.8 Research Judgment

How to Use This Page

Answer each question in 30–90 seconds. Start with the object and assumptions, state the key equation only if it clarifies the answer, then name one failure mode and one validation step.

1. Your backtest Sharpe is 3. What do you check first?
2. A feature has t=2.5 but no economic story. Keep it?
3. A feature has strong story but weak OOS evidence. Keep researching?
4. A model improves MSE but worsens net P&L. Which wins?
5. A signal works only in 2008 and 2020. What does that mean?
6. How many models did you test before your winner?
7. When is a complex model justified?
8. What would make you kill a live signal?
9. How do you explain a failed research project?
10. What makes research reproducible?

What a Strong Interviewer Hears

Senior interviewers care less about memorized equations and more about whether you know when evidence is insufficient, biased, non-actionable, or unstable.

Quick Check

Pick any three questions above and answer without using the words "because the model says so."